

BIS Sahayak (Bureau of Indian Standards Assistant)

Simple Project Guide & Complete Context • Smart India Hackathon 2026 (SIH26107)

Live Website: <https://sih2026-bis-assistant.vercel.app>

GitHub: github.com/PhoenixOP231/BIS_Sahayak_SIH

1. The Real-Life Problem: Why Did We Build This?

In India, millions of citizens buy everyday products like pressure cookers, drinking water bottles, electrical cables, helmets, and children's toys marked with an ISI stamp. However, many of these are dangerous counterfeits that cause household fires, cooker explosions, and health hazards. Meanwhile, small manufacturers (MSMEs) struggle to read 100-page dense government PDFs to find mandatory laboratory testing limits.

The Problem: Indian Standards are safety-critical and legally mandatory, but they are locked in complicated legal PDFs that regular citizens cannot understand and small businesses struggle to search.

2. What is BIS Sahayak?

BIS Sahayak (■■■■■■■ ■■■■■■) is an intelligent, 24/7 digital AI companion for product safety. Grounded in 21 official Bureau of Indian Standards documents, it answers questions instantly with two specialized modes:

- **Consumer Mode (For Citizens):** Plain language explanations, counterfeit warnings, buying checklists, and BIS Care App complaint instructions.
- **Industry Mode (For MSMEs & Engineers):** Exact numerical test tolerances (tensile stress, burst pressure, voltage), mandatory QCO orders, and routine test tables.

3. Step-by-Step User Journey

Consumer Walkthrough: A citizen visits the site, types *'Is my pressure cooker safe?'*, listens to the Hindi audio playback, and uses the **Verify ISI Mark** tool to check the CM/L license number against the BIS Care App.

Manufacturer Walkthrough: A steel mill owner selects **Industry Mode**, searches Fe 500D rebar limits, instantly receives exact mechanical parameters (500 MPa yield stress, 16% elongation), and clicks the citation chip to inspect full clause details.

4. Key Superpowers & Features

- **Zero-Hallucination AI (RAG):** Answers are strictly retrieved from authentic BIS standards, eliminating false answers.
- **4-Step ISI License Verifier:** Validates 7/8-digit CM/L license formats and guides users to official registry checks.
- **Standards Directory:** Searchable catalog of 21 standards across civil, electrical, safety, food, and toy sectors.
- **100% Bilingual & Voice Synthesis:** Fluent English & Devanagari Hindi with Web Speech API audio playback.

5. Why This Matters for SIH 2026

- **Citizens:** Protects families from hazardous, fake goods (exploding cookers, toxic toys, gas cylinder leaks).
- **MSMEs:** Boosts **'Make in India'** by making quality compliance fast, easy, and free of expensive consultancy fees.
- **Government:** Drives citizen engagement and downloads for the official **BIS Care Mobile App**.